MISSION UPDATE

NASA's Swift Telescope Ends Its 22-Year Mission


Artist’s concept

NASA's Swift space telescope ended its science mission on Oct. 20, after more than 22 years. When it detected a burst of gamma rays, Swift quickly turned its X-ray and ultraviolet/visible telescopes toward the same spot. That let astronomers follow the explosion and study how it changed over time. Swift could also alert other observatories, so they could watch the burst's fading afterglow.

High solar activity warmed and expanded Earth's upper atmosphere, increasing the drag on Swift. Its orbit sank until the spacecraft could no longer observe reliably. In its final weeks, it still studied hundreds of targets, including black holes and comets. When operations ended, Swift was about 275 kilometers above Earth. NASA expects it to re-enter the atmosphere in the coming weeks.

SOURCE / NASA Science: NASA's Swift Mission Ends Ahead of Re-Entry

OBSERVATIONAL ASTRONOMY

How Webb Sees New Stars Taking Shape

A close-up of the N159 star-forming cloud shows how stars shape their surroundings.



N159 is a huge cloud of gas and dust about 161,000 light-years away in the Large Magellanic Cloud, a nearby companion galaxy to the Milky Way. The region spans about 230 light-years. Inside it, cold hydrogen clouds collapse under gravity to form stars. The hottest young stars make nearby gas glow, creating bright bubbles and ridges in Webb's image. Their strong light and winds also push gas aside and shape the cloud.

Webb observes infrared light, which can reveal stars and structures hidden by dust. Its camera filters show different materials: orange marks carbon-rich molecules, blue shows hydrogen gas lit by young stars, and red shows cooler hydrogen. The colors are assigned to measurements from different filters, so they help researchers tell materials apart. These details help astronomers see where stars are forming and how their light and winds shape the surrounding cloud.

SOURCE / ESA/Webb: Stars brewing in a nebulous cauldron | Image credit: ESA/Webb, NASA, CSA